

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 3H nr. 12.8

$$r = 2 + \sin^2 4\theta$$

1. Domein

$$\text{dom } r = \mathbb{R}$$

$$p = 4\theta \Rightarrow \theta = \frac{\pi}{4}$$

$\Rightarrow$ we onderzoeken op het beperkt domein $\left[0, \frac{\pi}{4}\right]$

2. Tekenonderzoek van $r(\theta)$

$$2 + \sin^2 4\theta = 0$$

$$\sin^2 4\theta = 0$$

$$\sin 4\theta = 0$$

$$4\theta = k\pi$$

$$\theta = \frac{k\pi}{4}$$

Maar omdat $\sin^2 4\theta \geq 0$ geldt:

$$r(\theta) = 2 + \sin^2 4\theta \geq 2$$

$$\Rightarrow r^{(-1)}(0) = \emptyset$$

θ	$\left \begin{array}{cc} 0 & \frac{\pi}{4} \end{array} \right.$
$r(\theta)$	$\left \begin{array}{cc} 2 & + \quad 2 \end{array} \right.$

3. Tekenonderzoek van $r'(\theta)$

$$r(\theta) = 2 + \sin^2 4\theta$$

$$r'(\theta) = 2 \sin 4\theta \cdot \cos 4\theta \cdot 4$$

$$r'(\theta) = 8 \sin 4\theta \cos 4\theta$$

$$8 \sin 4\theta \cos 4\theta = 0$$

$$\sin 4\theta = 0 \quad \text{of} \quad \cos 4\theta = 0$$

$$\sin 4\theta = 0 \Rightarrow 4\theta = k\pi \Rightarrow \theta = \frac{k\pi}{4}$$

$$\cos 4\theta = 0 \Rightarrow 4\theta = \frac{\pi}{2} + k\pi \Rightarrow \theta = \frac{\pi}{8} + \frac{k\pi}{4}$$

$$\Rightarrow r'^{(-1)}(0) = \left\{0, \frac{\pi}{8}, \frac{\pi}{4}\right\}$$

θ	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$
$r'(\theta)$	0	$+$	0
	0	$-$	0

Maximum:

$$r\left(\frac{\pi}{8}\right) = 2 + \sin^2\left(4 \cdot \frac{\pi}{8}\right)$$

$$r\left(\frac{\pi}{8}\right) = 2 + \sin^2\left(\frac{\pi}{2}\right)$$

$$r\left(\frac{\pi}{8}\right) = 2 + 1$$

$$\Rightarrow \max r\left(\frac{\pi}{8}\right) = 3$$

Minimum (randpunten):

$$r(0) = 2 + \sin^2(0)$$

$$r(0) = 2$$

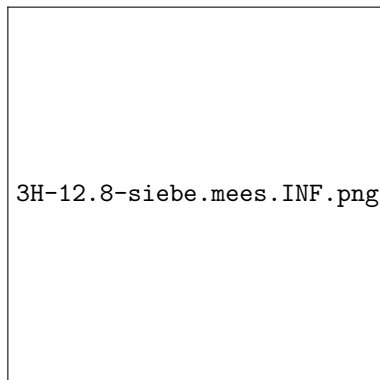
$$r\left(\frac{\pi}{4}\right) = 2 + \sin^2\left(4 \cdot \frac{\pi}{4}\right)$$

$$r\left(\frac{\pi}{4}\right) = 2 + \sin^2(\pi)$$

$$r\left(\frac{\pi}{4}\right) = 2$$

$$\Rightarrow \min r(0) = \min r\left(\frac{\pi}{4}\right) = 2$$

4. Schets



⇒ Samenvattende tabel

θ	0		$\frac{\pi}{8}$	$\frac{\pi}{4}$	
$r(\theta)$	+		+	+	
$r'(\theta)$	0	+	0	−	0
$r(\theta)$	$m(2)$	$\nearrow$	$M(3)$	$\searrow$	$m(2)$

References